

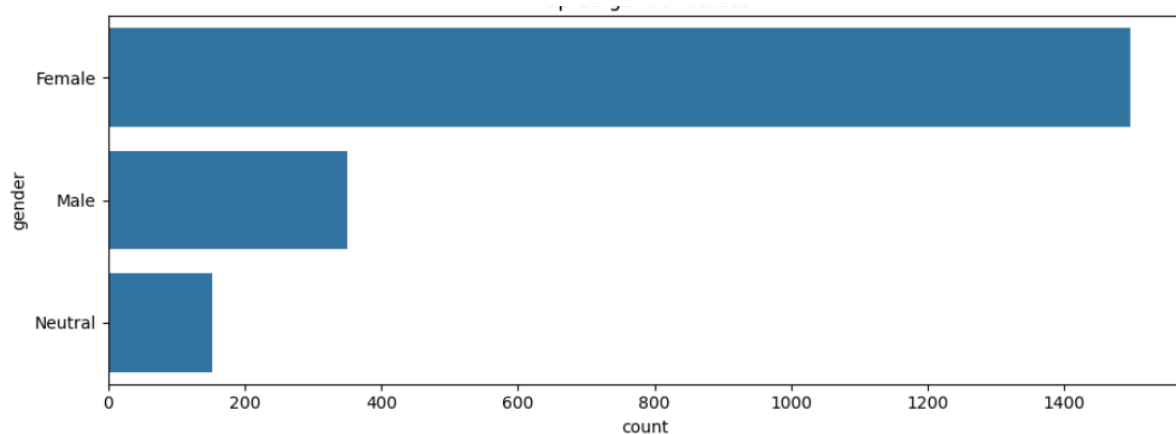
We are utilizing the iMaterialist dataset available on Hugging Face ([Marqo/iMaterialist](#)) as the foundation for our product catalog analysis.

The dataset provides a rich collection of fashion-related attributes, making it ideal for understanding product diversity, and visual characteristics within the apparel domain.

The iMaterialist dataset is very large(70gb+ and 721k+rows) so we have only used a extract of first 2000 rows present in it to perform this EDA

EDA on iMaterialist

Gender Balance



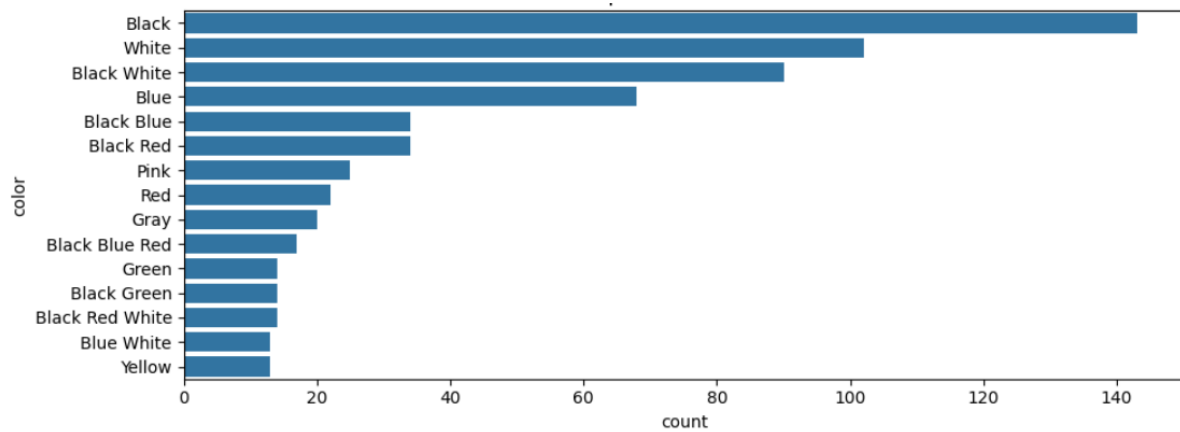
Female-oriented products dominate the dataset, while male and unisex categories account for a significantly smaller share.

The strong representation of female fashion items suggests that the dataset is primarily tailored toward women's apparel, which aligns with global retail trends — women's clothing consistently makes up the majority of fashion inventories and online product listings.

This gender imbalance is typical in fashion datasets, as women's categories tend to have greater product variety in terms of color, pattern, style, and seasonal trends.

The balanced presence of male and neutral items, however, ensures that the dataset still provides useful diversity for cross-gender product analysis and recommendation tasks.

Color distribution:

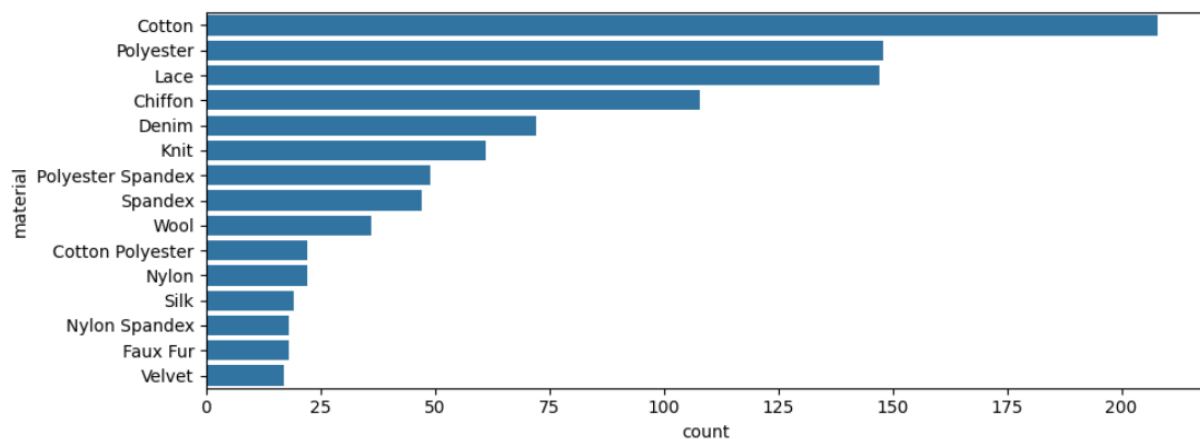


The most frequent colors are Black, White, and combinations like Black & White or Black & Blue.

Dark neutral tones such as black and white dominate because they are versatile and timeless, making them highly popular in both casual and formal wear.

This aligns with global fashion trends — black is considered universally flattering, and brands often offer multiple monochrome variations of the same design to appeal to wider audiences.

Material distribution:



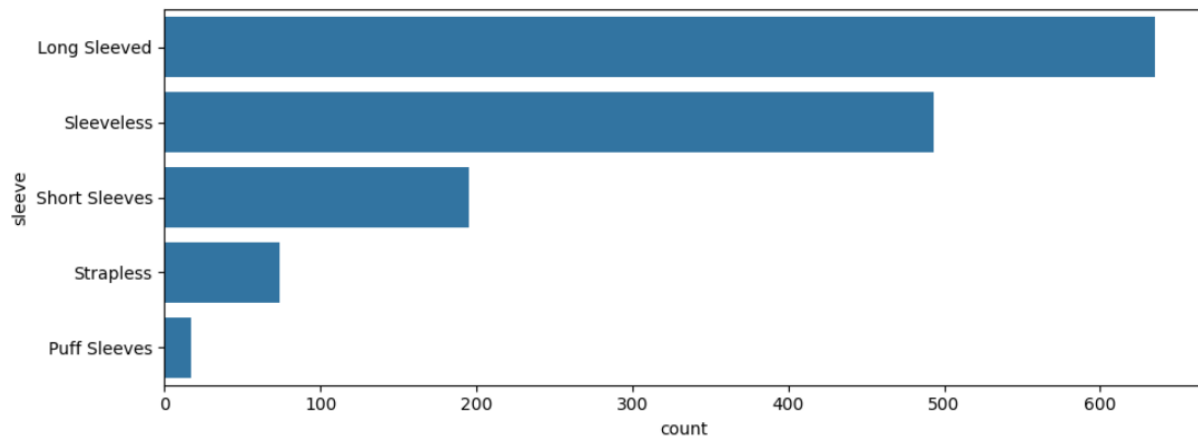
The most common materials are Cotton and Polyester, followed by Lace, Chiffon, and Denim.

Cotton and polyester are preferred due to their comfort, affordability, and suitability for mass production.

Lace and chiffon indicate a strong representation of feminine and dressy garments, suggesting that the dataset likely includes a high number of women's apparel items.

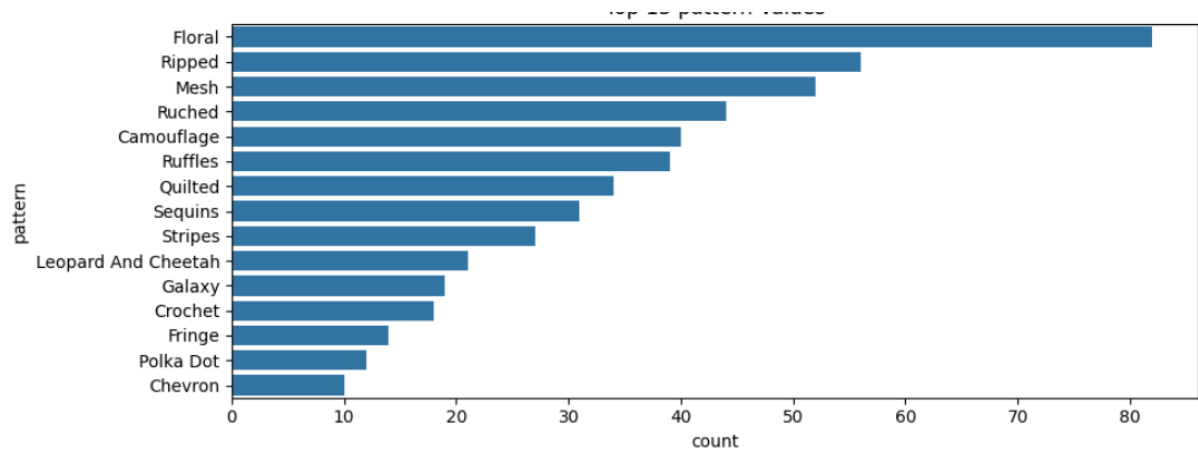
The mix of natural (cotton) and synthetic (polyester, spandex) fibers also shows balance between comfort and elasticity-based performance materials.

Sleeve Type distribution:



Most items have long sleeves, followed by sleeveless and short-sleeved designs.

pattern distribution:



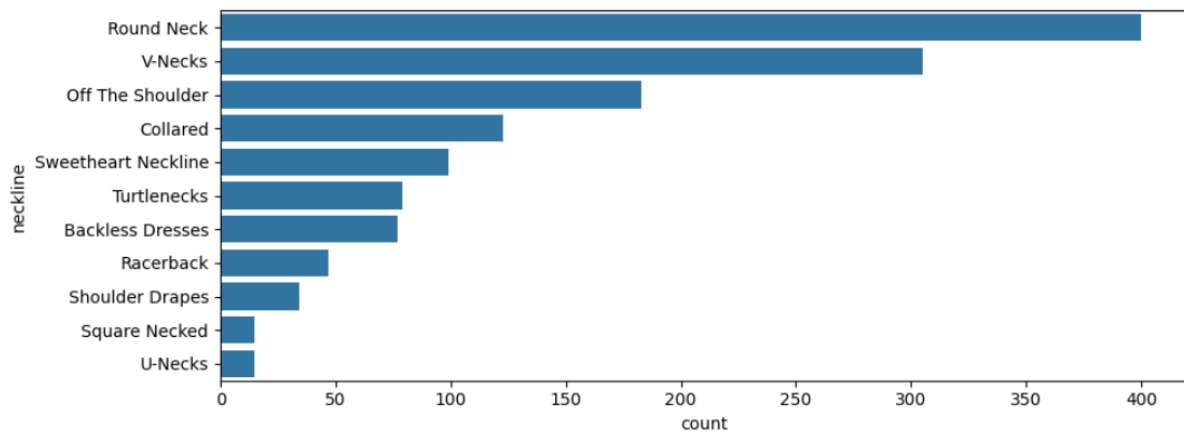
The most frequent patterns are Floral, Ripped, Mesh, and Ruched designs.

Floral patterns dominate as they are highly popular in women's fashion, particularly for dresses and blouses.

Ripped and ruched patterns indicate the inclusion of trendy and youth-oriented styles like ripped jeans or ruched tops.

This suggests the dataset captures a wide variety of modern and classic patterns, reflecting fast fashion diversity.

Neckline distribution:

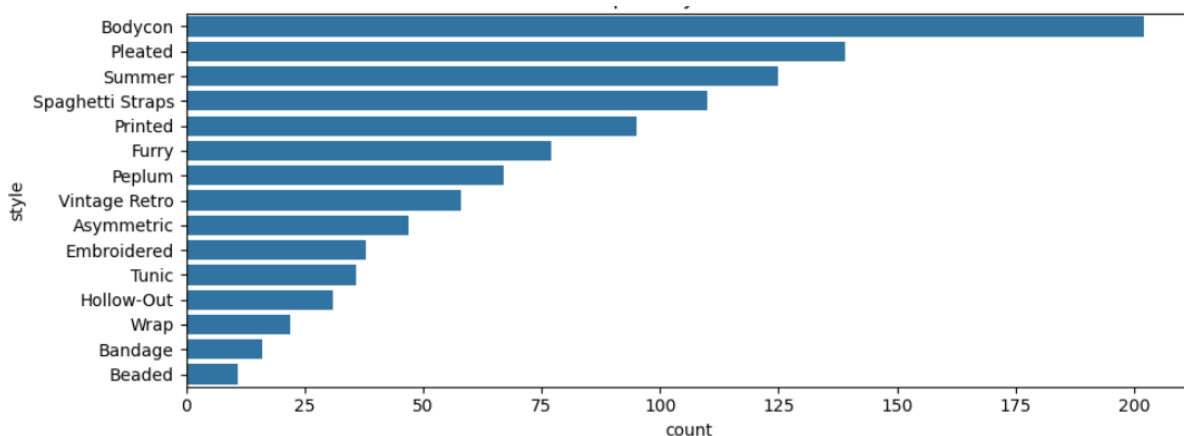


The leading neckline types are Round Neck and V-Neck, followed by Off-the-Shoulder and Collared styles.

Round and V-necks are the most common because they are simple, versatile, and work well across genders and occasions.

Off-the-shoulder and sweetheart necklines show a skew toward feminine designs, again consistent with fashion data where women's wear dominates

style distribution:



The dataset shows that Bodycon, Pleated, and Summer styles are the most prevalent, followed by Spaghetti Straps, Printed, and Furry designs.

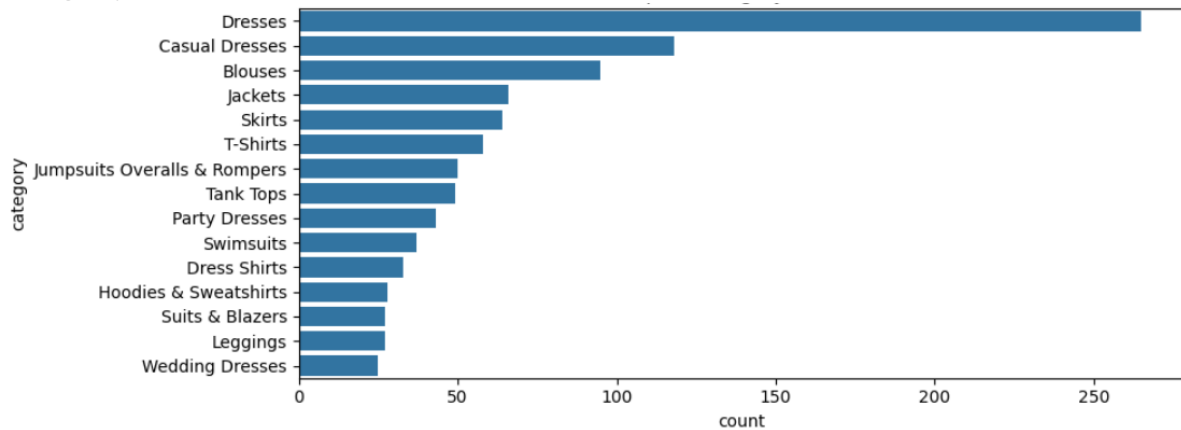
Less frequent styles include Wrap, Bandage, and Beaded, which represent niche or specialized fashion categories.

The dominance of Bodycon and Pleated styles suggests a strong emphasis on modern and form-fitting women's fashion, commonly seen in urban and western apparel markets.

The high frequency of Summer and Spaghetti Strap styles reflects the popularity of seasonal,

warm-weather clothing, which aligns with the overall gender distribution (female-dominated dataset).

Category distribution:



The highest number of items fall under Dresses, followed by Casual Dresses, Blouses, and Jackets.

Smaller but notable categories include Skirts, T-Shirts, Tank Tops, and Party Dresses, while more niche items like Wedding Dresses, Leggings, and Blazers appear less frequently.

The dominance of *Dresses* and *Casual Dresses* indicates that the dataset is heavily focused on women's apparel, which aligns with real-world fashion trends — women's categories typically occupy the largest share in online clothing catalogs.

This suggests that the dataset primarily represents female-oriented fashion segments such as everyday wear, office wear, and occasion dresses.

The presence of diverse categories like *Jackets*, *T-shirts*, and *Hoodies* also implies that it covers multi-seasonal clothing, suitable for both casual and semi-formal settings.

Overall, the distribution reflects a balanced yet female-majority dataset, ideal for studying product style, texture, and design variations across women's fashion types.

OTHER DATASET INFO

For the customer purchase dataset, we have decided to create a custom dataset tailored to our project requirements.

However, before doing so, we analysed several sample datasets available online to understand common structures and relevant features.

These reference datasets typically include columns such as customer ID, item ID, product information, mode of purchase, return reason, payment method (debit or credit card), geographical location, and product size.

We plan to incorporate similar attributes into our customized dataset to ensure it captures realistic and comprehensive customer purchase behaviour.